

CANNON AFB, NM, USA

WMO: 722686

Lat: 34.383N

Lon: 103.317W

Elev: 4295

StdP: 12.55

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 23008

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.9	18.3	-1.2	6.0	38.6	2.4	7.3	39.6	33.4	47.1	30.1	51.5	9.9	20	0.601

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.8	98.7	63.1	95.8	63.3	93.3	63.8	71.1	79.9	69.6	80.0	68.5	80.3	13.3	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.7	128.2	72.3	67.7	119.7	70.9	66.0	112.5	70.5	37.7	80.3	36.4	79.0	35.3	79.5	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
30.4	26.4	23.5	DB	6.8	102.8	5.9	3.0	2.6	105.0	-0.8	106.8	-4.1	108.5	-8.4	110.7
			WB	6.3	73.8	5.8	2.8	2.1	75.9	-1.3	77.6	-4.6	79.2	-8.8	81.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.1	39.2	43.7	50.4	57.6	66.3	75.8	78.5	77.2	71.0	59.2	48.5	40.3	(6)
	DBStd	16.00	9.21	9.67	8.94	8.39	7.57	5.19	4.12	4.04	6.26	8.28	8.84	9.23	(7)
	HDD50	1167	348	214	105	31	2	0	0	0	0	23	128	316	(8)
	HDD65	3658	800	596	453	241	75	2	0	0	21	210	494	766	(9)
	CDD50	4478	13	38	119	258	509	775	883	843	632	309	84	15	(10)
	CDD65	1492	0	0	1	18	116	327	418	379	203	30	0	0	(11)
	CDH74	17894	2	21	150	602	1812	4039	4647	3919	2129	524	48	1	(12)
	CDH80	8794	0	2	20	155	810	2200	2507	2014	945	139	2	0	(13)
(14) Wind	WSAvg	11.3	11.8	12.4	12.7	13.5	12.2	11.4	9.7	9.1	9.6	10.7	11.2	11.7	(14)
(15) Precipitation	PrecAvg	16.3	0.5	0.3	0.8	1.0	1.9	1.9	2.4	2.4	2.1	1.7	0.5	0.7	(15)
	PrecMax	27.8	1.4	1.3	2.9	4.5	10.4	4.4	8.5	6.2	5.5	6.9	1.5	3.3	(16)
	PrecMin	6.1	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.0	0.0	0.2	0.0	0.0	(17)
	PrecStd	5.0	0.5	0.4	0.7	1.0	2.1	1.3	1.7	1.4	1.3	1.9	0.5	0.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	71.3	77.0	82.7	88.5	96.8	103.5	101.3	98.9	96.6	88.1	78.9	71.5	(19)
		MCWB	46.7	49.9	51.8	54.2	58.6	61.8	63.8	64.5	62.8	59.0	51.8	47.3	(20)
	2%	DB	65.8	72.1	78.5	84.2	92.6	99.3	98.0	95.6	92.6	83.9	74.7	65.9	(21)
		MCWB	45.1	47.4	49.9	52.5	57.3	61.4	64.9	65.4	62.5	57.6	50.8	46.2	(22)
	5%	DB	61.1	67.6	74.6	81.1	88.8	95.7	95.1	93.2	89.6	80.6	70.0	61.7	(23)
		MCWB	42.9	45.8	48.9	52.1	56.4	60.9	65.6	65.6	62.5	56.7	49.5	43.8	(24)
	10%	DB	56.3	62.6	69.9	77.2	84.4	92.1	92.3	90.7	85.9	75.8	64.6	56.5	(25)
		MCWB	40.8	44.0	47.5	51.4	56.7	62.4	65.8	65.7	62.6	55.5	47.6	41.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	49.0	52.8	56.5	63.0	68.1	71.3	73.4	73.7	70.2	68.3	57.6	51.1	(27)
		MCDB	65.9	67.4	66.9	67.0	75.0	79.8	80.5	78.6	78.4	71.0	63.9	63.3	(28)
	2%	WB	46.3	49.6	53.1	58.9	64.9	69.1	70.9	71.3	68.2	64.1	54.7	48.0	(29)
		MCDB	62.7	66.9	68.7	67.0	75.8	80.2	81.2	79.1	77.6	71.0	64.3	61.9	(30)
	5%	WB	43.9	47.0	51.0	55.8	63.0	67.8	69.4	69.6	66.8	61.2	51.9	45.4	(31)
		MCDB	59.5	65.4	67.5	70.1	74.6	80.2	81.0	79.8	77.4	70.3	64.5	59.6	(32)
	10%	WB	41.4	44.7	49.1	53.6	60.9	66.4	68.4	68.4	65.5	58.7	49.7	42.7	(33)
		MCDB	55.0	60.8	66.9	71.7	74.1	80.0	81.3	80.4	76.9	69.9	62.8	55.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	26.5	28.4	29.5	29.9	28.5	28.6	25.8	25.3	25.7	26.6	27.0	25.9	(35)
		MCDBR	33.7	35.7	36.3	35.4	34.5	34.4	30.1	29.1	30.5	32.5	33.7	32.8	(36)
	5% WB	MCWBR	17.7	17.7	16.0	14.8	13.7	10.3	8.2	8.0	9.4	12.7	15.5	17.1	(37)
		MCWBR	31.7	33.3	32.1	30.4	29.1	27.1	24.3	23.9	24.1	23.8	27.8	29.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.247	0.256	0.283	0.308	0.322	0.361	0.373	0.359	0.329	0.297	0.270	0.254	(40)
		taud	2.604	2.594	2.513	2.437	2.456	2.382	2.396	2.416	2.472	2.540	2.549	2.579	(41)
	Ebn at Noon		313	320	316	310	305	291	287	290	296	299	300	304	(42)
		Edh at Noon	23	27	31	36	36	38	38	36	33	28	25	23	(43)
(44) All-Sky Solar Radiation	RadAvg		1063	1342	1741	2157	2366	2479	2336	2126	1815	1427	1118	921	(44)
	RadStd		84	108	124	125	141	117	107	101	108	139	95	58	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (11 neighbors)		+0.63	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+198	+113	

Nomenclature: See separate page